



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics Graphing

Do Now. Three different questions about the same moving car

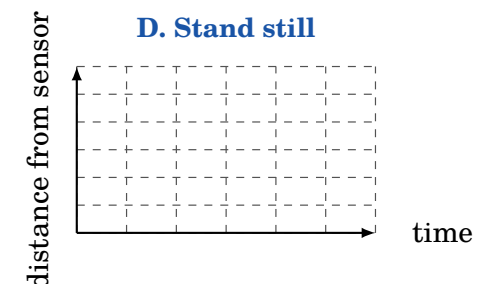
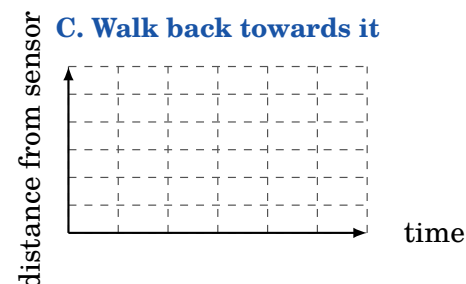
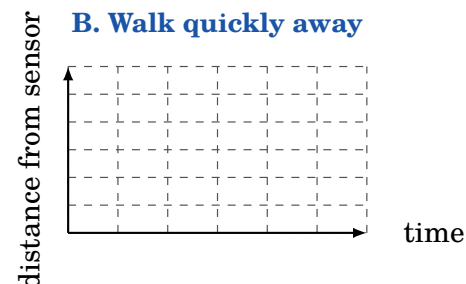
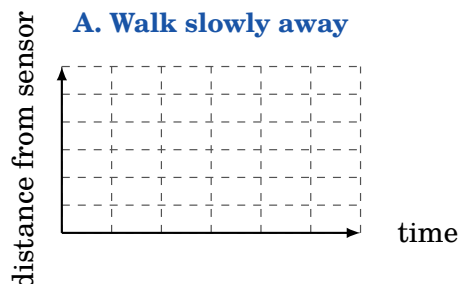
No graphs yet. Tick **one** box on each line.

Three separate questions, three separate answers. **Position** — *where is it?* (m). **Velocity** — *how fast, and which way?* (m/s). **Acceleration** — *how quickly is the velocity changing?* (m/s every second).

What somebody has told you about the car	position	velocity	acceleration
1. "It is 30 m past the school gate."			
2. "It is moving at 12 m/s, north."			
3. "It picks up another 2 m/s every second."			
4. "It is holding a steady 12 m/s." — which one is zero ?			
5. "It is reversing at a steady 3 m/s." — which one is negative ?			
6. "It is slowing from 20 m/s to a stop." — which one is negative ?			

Part 1. Your own motion, on the sensor

Sketch the *shape* of the line. There are no numbers on these axes on purpose.



What was different about the line...

between A and B? _____

between A and C? _____

and what did D look like? _____

About the third graph, before you ask. Every motion overleaf has an **acceleration-time** column, and in all six the line on it is **flat**. It is there so you have seen it — *nothing on this unit's test asks you to draw one*. Draw it, do not worry about it.

Name: _____

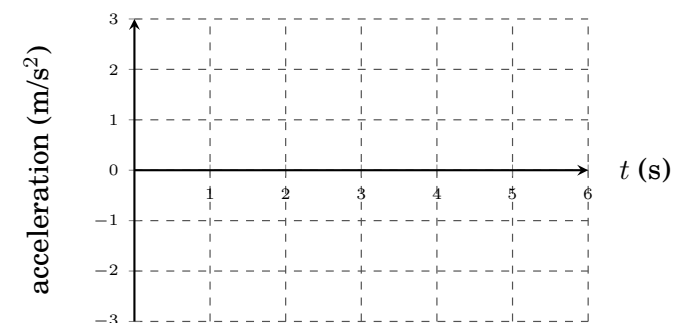
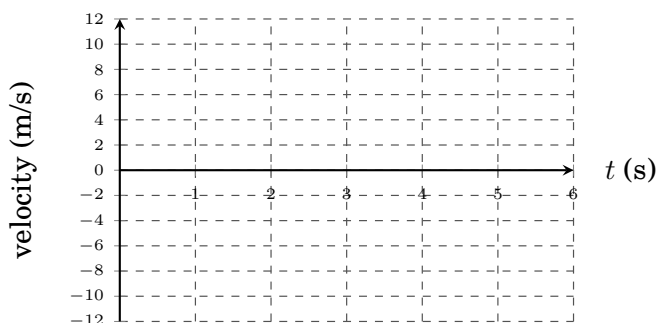
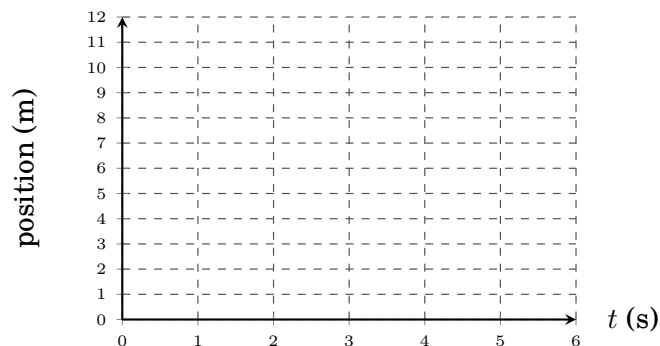
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Part 2 — The Basic 6. On the board is a moped that moves in six different ways. For each one, sketch the **three** graphs that describe it — position–time, velocity–time, acceleration–time. The axes are the *same* ones on screen, so copy what you see, and **write down the three numbers at the top of the screen first** — every line depends on them. *Straight, curved, or flat?* is the whole question; for velocity and acceleration, also ask whether the line sits **above or below** the time axis.

The Physics Classroom — Kinematic Graphing

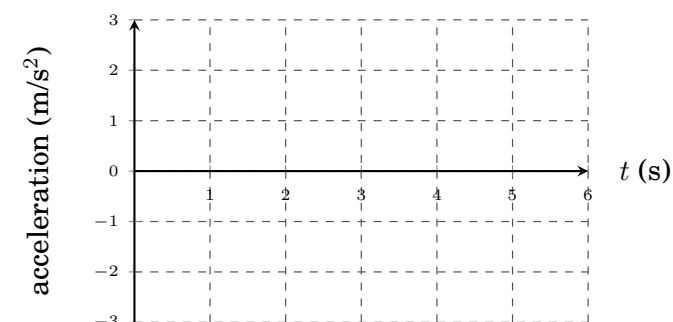
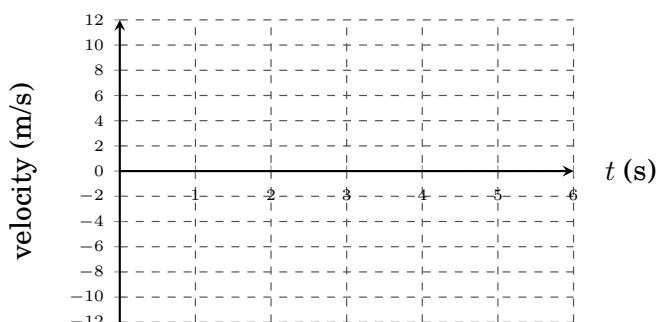
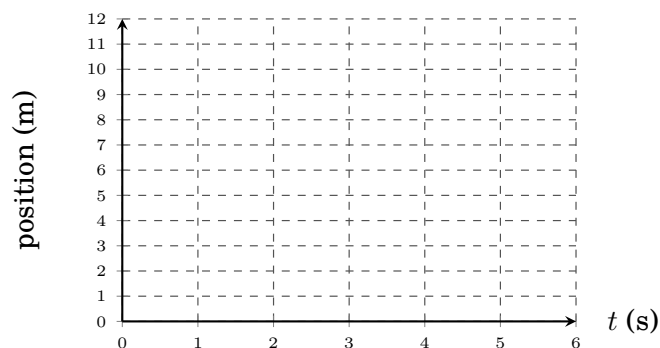
1. Move with Constant Speed in + Direction

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



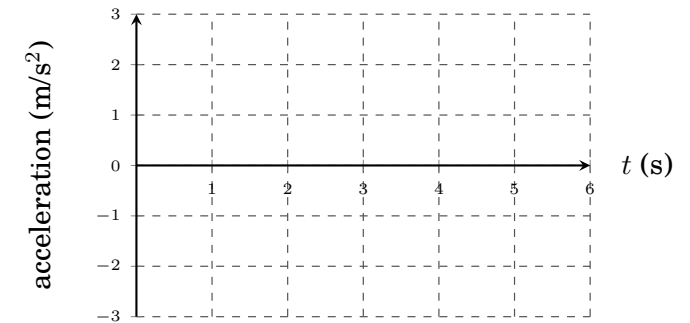
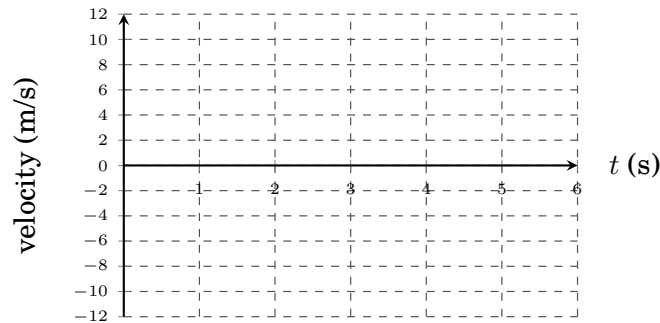
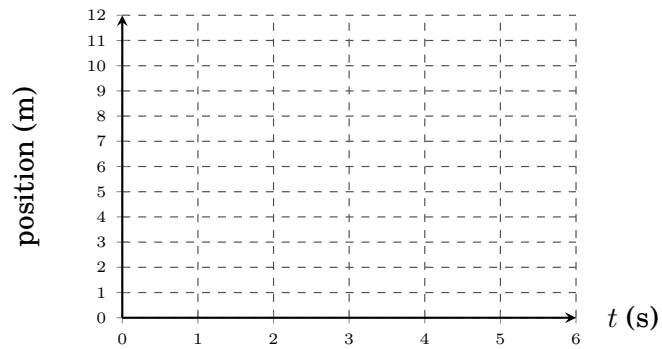
2. Move with Constant Speed in – Direction

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



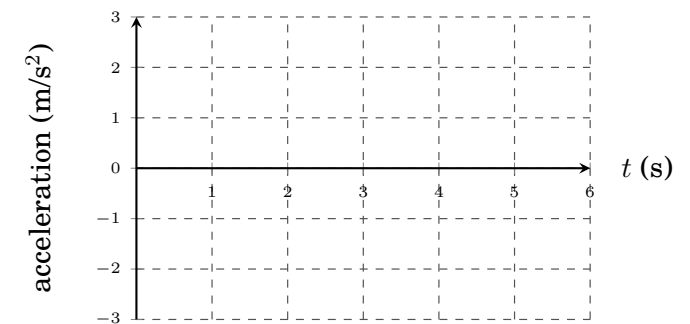
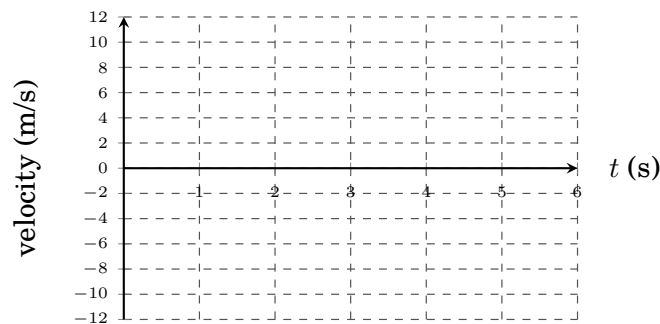
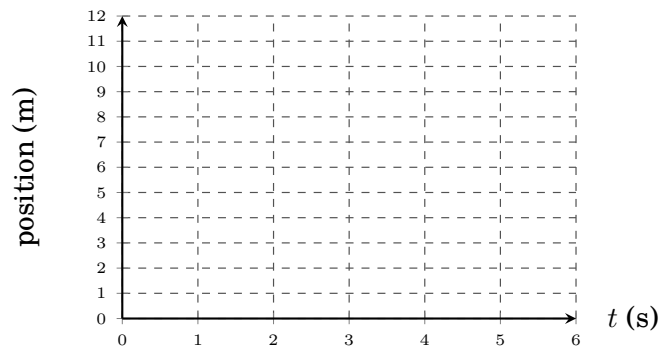
3. Move in + Direction; Speeding Up

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



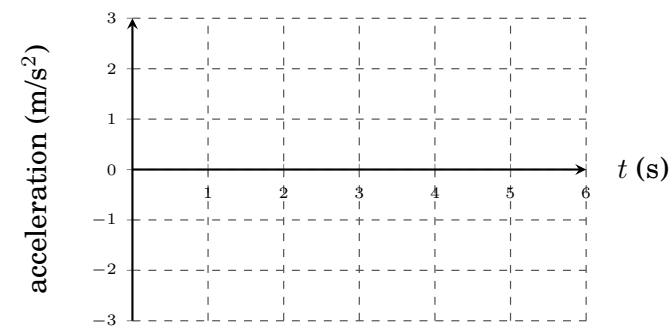
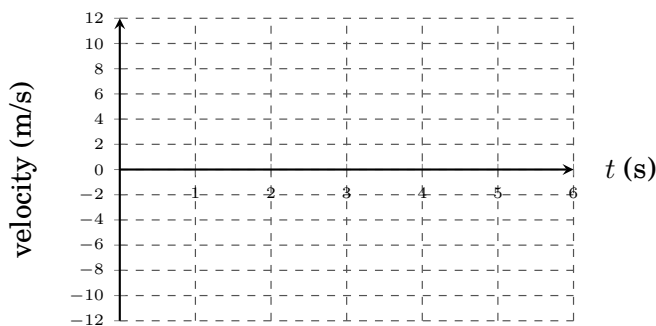
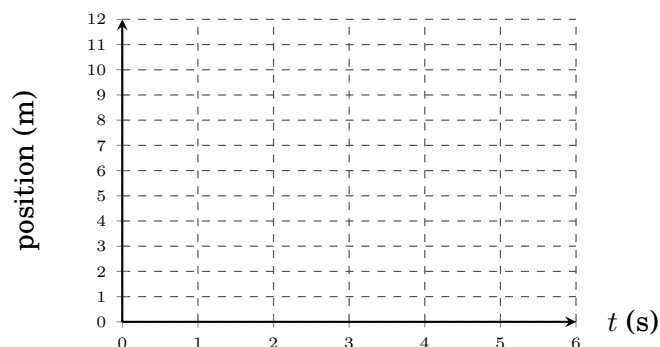
4. Move in + Direction; Slowing Down

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



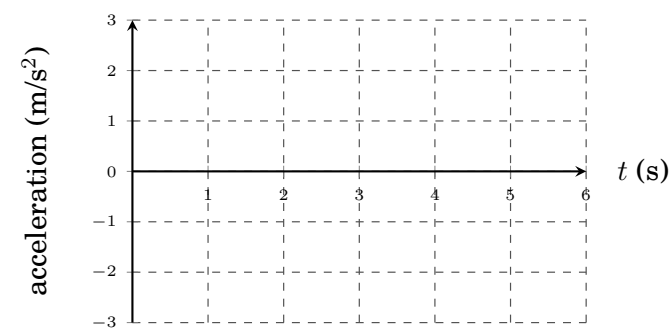
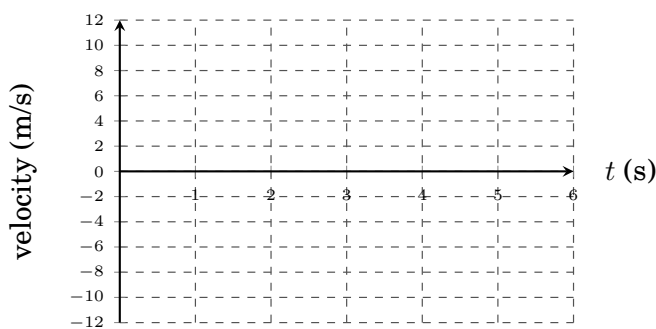
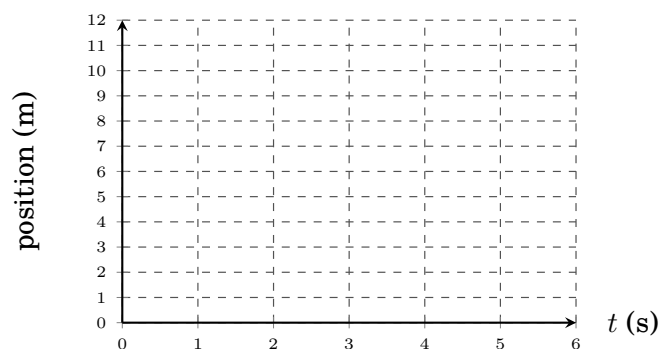
5. Move in – Direction; Speeding Up

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



6. Move in – Direction; Slowing Down

Initial velocity _____ m/s • Acceleration _____ m/s² • Maximum time _____ s



The one that catches everybody — and motions 5 and 6 are the proof

“Speeding up” does not mean positive acceleration. It is tempting to read a positive acceleration as *getting faster* and a negative one as *slowing down*. That is wrong. What decides it is whether the velocity and the acceleration point the **same way**: same sign as each other $\Rightarrow$ it **speeds up**; opposite signs $\Rightarrow$ it **slows down**.

So look at what you have just drawn. Motion **5** moves in the – direction and speeds up, and its acceleration is _____. Motion **6** moves in the – direction and slows down, and its acceleration is _____ — even though nothing about it is getting faster. **Read the two graphs together, never one alone.**